Week 2 Homework

Sigma notation review: Write the sum in expanded form:
a. $\sum_{i=1}^{5} \sqrt{i}$
b. $\sum_{i=1}^{n} i^{10}$
c. $\sum_{j=0}^{n-1} (-1)^j$
d. $\sum_{j=n}^{n+3} j^2$

Sigma notation review:Write the sum in sigma notation:
a. $1 + 2 + 3 + 4 + 5 + 6 + \cdots + 10$
b. $2 + 4 + 6 + 8 + \cdots + 2n$
c. $1 + x + x^2 + x^3 + \cdots + x^n$
d. $1 + 2x + 3x^2 + 4x^3 + \cdots + nx^{n-1}$

Factorial review: What is the factorial function? What notation is used for this function?

Factorial review: Calculate the following expressions using properties of the factorial function.
a. $\frac{7!}{3!}$
b. $\frac{8!}{4!4!}$
c. $\frac{10!}{9!}$
d. $\frac{n!}{(n-2)!}$

Sequence Review: Write the first five terms of the following sequences:
a. $a_1 = 1$, $a_{n+1} = \frac{1}{1+a_n}$
b. $b_1 = 0$, $b_2 = 1$, $b_n = b_{n-1} - b_{n-2}$
c. $c_1 = 1$, $c_n = c_{n-1} + n$
d. $d_1 = 1$, $d_n = 2d_{n-1}$

Sequence Review: Find a formula for the general term a_n of the given sequence:
a. $\{1, 4, 7, 10, 13 \cdots\}$
b. $\{-1, 2, -6, 24, -120 \cdots\}$
c. $\{1, 4, 9, 16, 25, \cdots\}$
d. $\{0, 1, 3, 7, 15, \cdots\}$